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FACULTY OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF MATHEMATICS

STA : 420 PROJECT IN STATISTICS

**Modeling Depression Severity among Caregivers in a Rural Setting
using Ordinal Logistic Regression**

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Bachelor of Science Statistics

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0.1 Declaration

We declare that this research project is our original work and has not been presented in any institution for any award or conferment of any degree.

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This project has been submitted for examination with my approval as the University Supervisor.

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0.2 Dedication

We dedicate this project to our parents, guardians, lecturers, relatives, and friends for their steadfast support during our university journey.

0.3 Acknowledgment

We wish to thank everyone who helped in diverse ways with ideas, insights, research materials and software(s) for the exploration of this project.

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0.4 Abbreviations and Acronyms

PHQ-9 - Patient Health Questionnaire-9

OLR - Ordinal Logistic Regression

LRT - Likelihood Ratio Test

EDA - Exploratory Data Analysis

KNBS - Kenya National Bureau of Statistics

OR - Odds Ratio

0.5 Definition of Terms

- Ordinal logistic regression.- s a statistical regression model used to analyze the relationship between one or more predictor variables and an ordinal response variable.
- Independent/predictor variables - Variables that are manipulated or changed to measure to the effect of that change on the outcome variable.
- Dependent/response variables -variables whose variation is being studied

0.6 Abstract

Support for cancer patients from their caregivers is crucial in terms of logistical, emotional, and physical care. However, the difficulties they encounter in rural areas can raise their vulnerability to depression. This abstract seeks to present a succinct overview of the incidence, contributing factors, and effects of depression in rural carers of cancer patients.

A comprehensive literature review was conducted, focusing on studies published between 2009 and 2023. Relevant databases including PubMed, PsycINFO, and Google Scholar were searched using keywords related to depression, caregivers, cancer patients, and rural settings. The selection criteria included studies that specifically examined the prevalence, risk factors, and impact of depression on caregivers in rural areas.

According to the analysis, rural caregivers for cancer patients frequently worry about depression. Increased levels of depression are a result of the particular difficulties that caregivers in rural locations confront, including restricted access to healthcare resources, and financial strain. This risk is further increased by the stress of providing care, which includes the physical difficulties, emotional anguish, and strained social interactions.

Due to the difficulties they face, caregivers for cancer patients in rural locations are more likely to experience depression. For the creation of focused therapies and support programs, it is imperative to comprehend the particular elements that lead to depression in this population. Future studies should investigate practical ways to meet the mental health requirements of rural caregivers, such as expanding social support systems, increasing access to mental healthcare, and offering caregiver education and training.

CHAPTER ONE

INTRODUCTION

1.1 Background Information

Depression is a mental health disorder characterized by persistent feelings of sadness, loss of interest or pleasure in activities, changes in appetite or sleep patterns, fatigue, difficulty concentrating, and thoughts of self-harm or suicide. It is a complex condition that can significantly impact individuals' emotional well-being, cognitive functioning, and overall quality of life.

Depressed individuals may have difficulty concentrating and making decisions, which can hinder their productivity in various aspects of life. Rehabilitation centers for depression have been established to provide specialized treatment facilities that provide comprehensive care for individuals experiencing moderate to severe depression.

1.2 Statement of the Problem

Researchers have been faced with the problem of identifying potential predictors of depression among cancer caregivers. The problem is to identify and to quantify the important variables linked to depression.

1.3 Objective of the Study

1.3.1 General objectives

To model depression severity among caregivers in a rural setting using ordinal logistic regression model.

1.3.2 Specific objectives

1. Identify significant predictors that contribute to depression severity among cancer caregivers in a rural setting
2. Estimate and interpret the coefficient of the factors leading to depression severity.
3. Determine and Interpret the odds ratio of the factors affecting depression severity.

1.4 Methodology Framework

Ordinal logistic regression is used in this study and R software is used for analysis. The data for the study was obtained from 700 participants. The independent variables were identified and were used to determine their relationship with the response variable.

1.5 Justification of the Study

Cancer caregiver depression is a serious mental health issue that can have a negative impact on both the caregivers and the cancer patients. This study identifies the unique elements leading to depression in these environments and informs alignment to specialized remedies inclusive of rural cancer caregiver support systems. Healthcare professionals, policymakers, and support groups can create targeted treatments that meet the unique needs and difficulties of caregivers in these circumstances by recognizing the predictors of depression giving access to mental health treatments, strengthening social support systems, and putting caregiver relief measures in place.

CHAPTER 2

LITERATURE REVIEW

In this chapter, a review of the existing research and scholarship related to mental health and its effects on a person's productivity as well as the interaction between an individual's cognitive, affective and somatic symptoms of depression is made. Key insights from previous research have immensely contributed to recent advances in mental health understanding. The following are some of the previous works.

A research study conducted by Martin et al. (2009), investigated whether different types of health promotion interventions in the workplace reduced depression and anxiety symptoms. A systematic review and meta-analysis of the literature was undertaken on workplace health promotion published during the period 1997-2007. Studies were considered eligible for inclusion if they evaluated the impact of an intervention using a valid indicator or specific measure of depression or anxiety symptoms. The standardized mean difference was calculated for each of the following three types of outcome measures; depression, anxiety and composite mental health. Altogether, 22 studies met the inclusion criteria, with a total sample size of 3409 employees post intervention, and 17 of the studies were included in the meta-analysis representing 20 interventions-control comparisons. The pooled results indicated small but positive overall effects of the interventions with respect to symptoms of depression and anxiety, but no effect on composite mental health measures. The interventions that included a direct focus on mental health had a comparable effect on depression and anxiety symptoms as did the interventions with an indirect focus on risk factors.

Jain et al. (2013) examined the burden of depression on work productivity in the US workforce. The Patient Health Questionnaire for depression severity and the Health and Work Performance Questionnaire and Work Productivity and Activity Impairment (WPAI) Questionnaire for absenteeism and presenteeism were used to survey 1051 full-time employees in February 2010. The results indicated that out of the 1051 employees with depression, 40.3% had no depressive symptoms at the time of the survey, 30.4% had mild depression, 15.8% had moderate depression, 7.8% had moderately severe depression, and 5.8% had severe depression. All levels of depression were associated with decreased work productivity as severity increased, presenteeism and absenteeism worsened.

In another study Serpa et al. (2014), investigated the major problems among Veter-

ans specifically anxiety, depression and suicidal ideation. Pre- and Post Mindfulness Based Stress Reduction (MBSR) questionnaires were used to collect data on MBSR's effectiveness among 79 veterans at an urban Veterans Health Administration medical facility and were used for comparison among individuals. A meditation analysis was conducted to determine whether changes in mindfulness were related to changes in the other outcomes for a period of 9 weekly sessions. The results indicated significant reductions in anxiety, depressions, and suicidal ideation after MBSR training as well as improved mental health functioning scores although pain intensity and physical health functionality did not register improvements.

Laing and Jones (2016) investigated the mediation effect of anxiety and depression on the relationship between perceived health-promoting workplace culture and presenteeism. Paper surveys were distributed to 4703 state employees. The variables that were investigated included symptoms of depression, anxiety, perceived workplace support for healthy living inclusive of physical activity and presenteeism. Correlational analysis was used to assess the relationships among culture, mental health and productivity. The results indicated that indirect effects of workplace culture on productivity, mediated by anxiety and depression symptoms were significant. Healthy living culture and anxiety were significantly negatively associated and anxiety and presenteeism were significantly positively associated.

Another study by Magtibay et al. (2017) assessed the efficacy of blended learning which was intended to decrease stress and burnout among nurses through the use of the Stress Management and Resiliency Training (SMART) program. Consistent with blended learning, participants chose the format that met their learning styles and goals; Web-based, independent reading or facilitated discussions. The end points of mindfulness, resilience, anxiety, stress, happiness, and burnout were measured at baseline, postintervention, and 3-month follow-up to examine within-group differences. The results indicated statistically significant, clinically meaningful decreases in anxiety, stress, and burnout and increases in resilience, happiness, and mindfulness.

Levis et al. (2019), evaluate the accuracy of the Patient Health Questionnaire-9 (PHQ-9) for screening major depression in adults using individual participant data meta-analysis. The study, which was conducted by the DEPRESSion Screening Data (DEPRESSD) Collaboration, analyzed data from 58 studies, with a total of 17 357 participants. It was found that combined sensitivity and specificity was maximized at a cut-off score of 10 or above among studies using a semi-structured interview.

Secondly across cut-off scores of 5-15, sensitivity with semi-structured interviews was 5%-22% higher than for fully structured interviews.

Leung et al. (2020) investigated the factor structure, reliability, and construct validity of the Patient Health Questionnaire-9 (PHQ-9) in a community sample of 10 933 Chinese adolescents. Confirmatory factor analysis (CFA) was used to examine the one-factor structure of the PHQ-9, and multiple-group CFAs were performed to test for measurement invariance across gender and age subgroups. Internal consistency reliability was evaluated using Cronbach alpha, and construct validity was assessed using Pearson correlation coefficients between PHQ-9 scores and anxiety, self-esteem and perceived control. The results indicated that a one-factor model with three pairs of items correlations fitted the PHQ-9 data well and that measurement invariances by age and gender were supported. The PHQ-9 was also strongly correlated with anxiety, self-esteem and perceived control.

In a recent study, Levis et al. (2020), a meta-analysis of 44 studies was conducted to compare the sensitivity and specificity of the PHQ-2 alone and combined with PHQ-9. The Individual participant data were obtained from 100 eligible studies comprising of 44 318 participants, of which 4572 had major depression. The authors found that the combination of PHQ-2 followed by PHQ-9 had similar sensitivity but higher specificity compared with PHQ-9.

In a more recent study, Lister et al. (2023) conducted a qualitative study that investigated the barriers and enablers to mental well being and academic success that students experienced in distance learning. Narrative inquiry was used whereby students shared their own stories and tutors shared stories about the students they had supported. In total, 60 themes were identified, consisting of 155 sub-themes and 783 coded references. The barriers and enablers were then clustered into three overall categories; study-related barriers/enablers, skills-related barriers/enablers and environmental barriers/enablers, and 10 sub-categories, curriculum, tuition, assessment, social skills, self-management, study skills, spaces, systems, people and life. It was found that barriers and enablers were existent in the majority of themes, implying that different aspects of study could be both barriers or enablers for mental well being and study success, depending on who was experiencing them and how, for example, assessment feedback was a barrier for students when it was negative, but was an enabler when it was constructive and took place in an environment of trust.

In summary, the past research works on mental health have been conducted using

Meta-analysis, Qualitative techniques, CFA and Narrative inquiry. In this study, the ordinal logistic regression model is used to examine the depression severity of cancer caregivers in a rural setting.

CHAPTER THREE

ORDINAL LOGISTIC REGRESSION

3.1 Introduction

This chapter presents the theory of the ordinal logistic regression model.

3.2 Ordinal Logistic Regression

Ordinal logistic regression is a statistical method utilized to analyze the association between a categorical with natural ordering and one or more explanatory variables. The explanatory variables can be either continuous or categorical in nature. Unlike logistic regression, which deals with binary responses, ordinal logistic regression extends this concept to handle responses with multiple ordered levels (Agresti, 2019).

3.3 Assumptions of the model:

For the results of ordinal logistic to hold, certain assumptions have to be met. The assumptions associated with Ordinal Logistic Regression are:

- **Proportional Odds Assumption:**
The proportional odds assumption states that the relationship between the independent variables and the cumulative odds of being in a higher category remains constant across all levels of the dependent variable. This assumption implies that the coefficients associated with the independent variables are the same across different levels of the dependent variable.
- **Independence of Observations:**
Each observation or data point is independent of others. This assumption requires that the observations are not influenced by each other, meaning that the outcome of one observation does not affect the outcome of another.
- **Ordinality of the Dependent Variable:**
Ordinal logistic regression assumes that the dependent variable is measured on an ordinal scale. This means that the categories of the dependent variable have a natural order and the distance between categories is not necessarily equal.
- **Absence of Perfect Separation:**
Perfect separation occurs when there is a combination of independent variable

values that perfectly predicts the outcome category. In such cases, the model cannot be estimated or interpreted properly.

- Adequate Sample Size:

Having an adequate sample size is important for reliable results. The sample size should be large enough to ensure stable parameter estimates and valid statistical inference. Small sample sizes can lead to imprecise estimates and low statistical power.

3.4 The Model

Let $\mathbf{Y}$ be an ordinal outcome with j categories, $j \geq 2$.

Suppose there are p covariates $X_1, X_2, \dots, X_p$

Then $P(Y \leq j)$ is the cumulative probability of $\mathbf{Y}$ less than or equal to a specific category $j = 1, \dots, j - 1$.

The odds of being less than or equal a particular category can be defined as $\frac{P(Y \leq j)}{P(Y > j)}$
The ordinal logistic regression model is defined as follows:

$$\text{logit}(P(Y \leq j)) = \alpha_j + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p \quad (3.1)$$

There exists three parameterizations of the ordinal logistic regression model

1. model 1:

$$\log \frac{P(Y \leq j)}{P(Y > j)} = \alpha_j - \beta_1 X_1 - \beta_2 X_2 - \dots - \beta_p X_p \quad (3.2)$$

2. model 2:

$$\log \frac{P(Y \leq j)}{P(Y > j)} = \alpha_j + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p \quad (3.3)$$

3. model 3:

$$\log \frac{P(Y \leq j)}{P(Y \leq j-1)} = \alpha_j - \beta_1 X_1 - \beta_2 X_2 - \dots - \beta_p X_p \quad (3.4)$$

where:

- $P(Y \leq j)$ is the cumulative probability of the response variable Y being in category j or lower,

- α_j is the intercept parameter specific to category j ,
- $\beta_1, \beta_2, \dots, \beta_p$ are the coefficients associated with the independent variables $X_1, X_2, \dots, X_p$, respectively.

In the re-parametrized model 1, a negative sign is incorporated in the coefficients to establish a direct correspondence between the slope and the ranking. Therefore, a positive coefficient in the model indicates that as the value of the explanatory variable increases, the likelihood of a higher ranking also increases. This re-parametrization helps in interpreting the coefficients in a more intuitive manner, as a positive coefficient signifies a positive effect on the response variable, aligning with our expectation that higher values of the explanatory variable lead to higher rankings.

3.5 Interpretation of model Coefficients

- The intercept parameter α_j represents the log-odds of the response variable being in category j or lower when all independent variables are zero.
- The coefficient β_k associated with independent variable X_k represents the change in the log-odds of the response variable for a one-unit increase in X_k , while holding all other variables constant.

To obtain the odds ratio, exponentiate the coefficient:

$$\text{Odds Ratio} = \exp(\beta_k)$$

The odds ratio indicates how the odds of being in a higher category change with a one-unit increase in the corresponding independent variable, assuming the other variables are held constant.

3.6 Prediction of New Observations

The predicted probabilities and proportional odds can be obtained for ordinal logistic regression. Using model 2,

$$\log \frac{P(Y \leq j)}{P(Y > j)} = \alpha_j + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p \quad (3.5)$$

the predicted probabilities can be calculated as

$$\log \frac{P(Y \leq j)}{P(Y > j)} = \frac{e^{\alpha_j + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p}}{1 + e^{\alpha_j + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p}} \quad (3.6)$$

When the assumption of proportional odds holds, the predicted probabilities derived from the model will closely align with the observed proportions.

3.7 Estimation of the Model Parameters

Suppose,

$$E[Y_i] = \pi_i \quad (3.7)$$

and

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} = \mathbf{x}_i^T \boldsymbol{\beta}, \quad (3.8)$$

where $\mathbf{x}_i = (1, x_{i1}, \dots, x_{ip})'$ and

$$\boldsymbol{\beta} = (\beta_0, \dots, \beta_p)'$$

Therefore,

$$E[Y_i] = \frac{e^{\mathbf{x}_i^T \boldsymbol{\beta}}}{1 + e^{\mathbf{x}_i^T \boldsymbol{\beta}}}$$

The likelihood for n observations then is;

$$L = \prod_{i=1}^n \frac{\exp(\mathbf{x}_i^T \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_i^T \boldsymbol{\beta}) (\prod_{i=1}^n (1 - \exp(\mathbf{x}_i^T \boldsymbol{\beta})))^{n - \sum_{i=1}^n Y_i}} \quad (3.9)$$

and the log-likelihood is

$$l = \sum_{i=1}^n Y_i \log \left(\frac{\exp(\mathbf{x}_i^T \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_i^T \boldsymbol{\beta})} \right) + (1 - Y_i) \log \left(1 - \frac{\exp(\mathbf{x}_i^T \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_i^T \boldsymbol{\beta})} \right). \quad (3.10)$$

3.8 Confidence Intervals for the Coefficients and the Odds Ratios

In the model

$$\log \frac{P(Y \leq j)}{P(Y > j)} = \alpha_j + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$$

a $(1 - \alpha/2) \times 100\%$ confidence interval for β_j , where $j = 1, \dots, p$, can easily be calculated as

$$\hat{\beta}_j \pm Z_{1-\alpha/2} \cdot \text{se}(\hat{\beta}_j).$$

The $(1 - \alpha/2) \times 100\%$ confidence interval for the odds ratio over a one unit change in x_j is given by

$$\left(\exp(\hat{\beta}_j - Z_{1-\alpha/2} \cdot \text{se}(\hat{\beta}_j)), \exp(\hat{\beta}_j + Z_{1-\alpha/2} \cdot \text{se}(\hat{\beta}_j)) \right).$$

3.9 Hypothesis Testing Likelihood Ratio Test

The likelihood ratio test (LRT) statistic calculates the ratio between the likelihood at the hypothesized parameter values and the likelihood of the data at the maximum likelihood estimate(s). This statistic is derived from the following formula: For ordinal logistic regression, the likelihood ratio test (LRT) statistic can be calculated as:

$$LR = -2 \log \left(\frac{L(\text{at } H_0)}{L(\text{at MLE}(s))} \right) = -2l(H_0) + 2l(\text{MLE}) \quad (3.11)$$

For large n , the LRT statistic follows a chi-square distribution with degrees of freedom equal to the number of parameters being estimated.

For a given binary outcome, if the hypothesis is $H_0 : p = p_0$ versus $H_A : p \neq p_0$, the log-likelihoods are:

$$\begin{aligned} l(H_0) &= \frac{1}{n} \sum_{i=1}^n Y_i \log(p_0) + \left(1 - \frac{1}{n} \sum_{i=1}^n Y_i \right) \log(1 - p_0) \\ l(\text{MLE}) &= \frac{1}{n} \sum_{i=1}^n Y_i \log(\hat{p}) + \left(1 - \frac{1}{n} \sum_{i=1}^n Y_i \right) \log(1 - \hat{p}) \end{aligned}$$

where Y_i represents the binary outcome variable, p_0 is the hypothesized probability, and $\hat{p}$ is the maximum likelihood estimate for the probability.

The LRT statistic is given by:

$$LR = -2 \left(\frac{1}{n} \sum_{i=1}^n Y_i \log \left(\frac{p_0}{\hat{p}} \right) + \left(1 - \frac{1}{n} \sum_{i=1}^n Y_i \right) \log((1 - p_0)(1 - \hat{p})) \right) \quad (3.12)$$

where $LR \sim \chi_1^2$.

CHAPTER FOUR

DATA ANALYSIS AND RESULTS

4.1 Introduction

In this chapter, we discuss the sources of our data and the process of the preparation. We also explore the characteristics of data and finally we analyze and interpret the results.

4.2 Data for the Study

The data utilized in this study is Depression among cancer caregivers in a rural setting dataset published on figshare repository(Kagawa et al., 2021). Trained research assistants (RA) with training in counseling and Responsible Conduct of Research (RCR), administered the translated 30-minute-long questionnaire (written form). This questionnaire captured the participants' sociodemographic characteristics; age, sex, marital status, level of education, employment status, religion, and monthly income. Depression was assessed using the patient health questionnaire. All participants provided written consent to participate in the study. The sample size of 700 was calculated using the Kish-Leslie formula, basing on an estimated prevalence of 38.2% among caregivers of cancer patients, level of attrition of 10%, 95% confidence interval and margin of error 5%.

Caregivers of cancer patients to participate in the study by convenience sampling based on the caregivers present during the time of data collection - in the evening (convenience time for most caregivers) when the caregivers were not so occupied with caregiving activities. Caregivers who were already receiving mental health care for depression (as reported by the caregiver) were excluded from the study. Because the study was screening for individuals who need treatment and those were already on treatment.

In this study, we utilized the "Age of participant" variable which represents the age of the caregiver who responded to the questionnaire, "Age of patient" variable, "Household income" which represents the amount of money the household earns on a monthly basis. The dependent variable (depression) was measured on ordered four levels; None=1, mild=2, moderate=3, severe=4 and profound=5.

4.3 Data Preparation

```

str(dataset)

## 'data.frame': 700 obs. of  9 variables:
##  $ age_participant: num  54 34 44 46 44 39 55 39 60 39 ...
##  $ age_patient    : num  51 53 25 38 25 32 21 23 58 40 ...
##  $ sex_participant: int   1 1 0 1 0 0 0 0 0 0 ...
##  $ sex_patient    : int   0 1 0 1 1 0 1 0 0 1 ...
##  $ marital_status : chr  "single" "single" "divorced" "married" ...
##  $ education      : chr  "secondary" "tertiary" "secondary" "primary" ...
##  $ income_ksh     : num  69596 182616 42987 1748831 13415 ...
##  $ social_support : int   0 1 0 0 0 1 0 0 0 0 ...
##  $ depression     : Factor w/ 5 levels "1","2","3","4",...: 2 1 2 2 1 2 2 1 1 5 ...

head(dataset, 10)

##      age_participant age_patient sex_participant sex_patient marital_status
## 1             54           51             1             0          single
## 2             34           53             1             1          single
## 3             44           25             0             0        divorced
## 4             46           38             1             1          married
## 5             44           25             0             1          married
## 6             39           32             0             0          single
## 7             55           21             0             1          single
## 8             39           23             0             0        divorced
## 9             60           58             0             0          single
## 10            39           40             0             1        divorced
##      education income_ksh social_support depression
## 1 secondary      69596             0             2
## 2 tertiary      182616             1             1
## 3 secondary      42987             0             2
## 4 primary      1748831             0             2
## 5 secondary      13415             0             1
## 6 tertiary      1474274             1             2
## 7 secondary      288196             0             2
## 8 tertiary       90785             0             1
## 9 primary       214242             0             1
## 10 primary       155082             0             5

View(dataset)

```



```

#data set preparation

# education levels
education_mapping <- c("primary" = 1, "secondary" = 2, "tertiary" = 3)

# marital status levels
marital_status_mapping <- c("married" = 1, "single" = 2, "divorced" = 3)

# Mutate the 'education' and 'marital_status' columns with numeric values and also
dataset <- dataset %>%
  mutate(education = education_mapping[education],
         marital_status = marital_status_mapping[marital_status],
         income_ksh=income_ksh/100000)

View(dataset)
df<-as.data.frame(dataset)
class(df)

## [1] "data.frame"

names(df)

## [1] "age_participant" "age_patient"      "sex_participant" "sex_patient"
## [5] "marital_status"  "education"         "income_ksh"      "social_support"
## [9] "depression"

```

4.4 Exploratory Data Analysis

The data set was subjected to exploratory data analysis with the objective to gain a better understanding of it. All the analysis in this study were done on R software.

```

# Summary statistics
summary(df)

## age_participant age_patient sex_participant sex_patient
## Min. :10.00 Min. : -22.00 Min. :0.0000 Min. :0.00
## 1st Qu.:33.00 1st Qu.: 31.00 1st Qu.:0.0000 1st Qu.:0.00

```

```
## Median :40.00 Median : 45.00 Median :1.0000 Median :1.00
## Mean :39.66 Mean : 44.46 Mean :0.5057 Mean :0.53
## 3rd Qu.:46.00 3rd Qu.: 58.00 3rd Qu.:1.0000 3rd Qu.:1.00
## Max. :72.00 Max. :115.00 Max. :1.0000 Max. :1.00
## marital_status education income_ksh social_support depression
## Min. :1.000 Min. :1.000 Min. : 0.0676 Min. :0.00 1:141
## 1st Qu.:1.000 1st Qu.:1.000 1st Qu.: 0.7426 1st Qu.:0.00 2:240
## Median :2.000 Median :2.000 Median : 1.5294 Median :0.00 3:161
## Mean :1.764 Mean :1.897 Mean : 2.6717 Mean :0.28 4: 99
## 3rd Qu.:2.000 3rd Qu.:2.000 3rd Qu.: 3.0880 3rd Qu.:1.00 5: 59
## Max. :3.000 Max. :3.000 Max. :38.9233 Max. :1.00

str(df)

## 'data.frame': 700 obs. of 9 variables:
## $ age_participant: num 54 34 44 46 44 39 55 39 60 39 ...
## $ age_patient : num 51 53 25 38 25 32 21 23 58 40 ...
## $ sex_participant: int 1 1 0 1 0 0 0 0 0 0 ...
## $ sex_patient : int 0 1 0 1 1 0 1 0 0 1 ...
## $ marital_status : Named num 2 2 3 1 1 2 2 3 2 3 ...
## ..- attr(*, "names")= chr [1:700] "single" "single" "divorced" "married" ...
## $ education : Named num 2 3 2 1 2 3 2 3 1 1 ...
## ..- attr(*, "names")= chr [1:700] "secondary" "tertiary" "secondary" "primary" ...
## $ income_ksh : num 0.696 1.826 0.43 17.488 0.134 ...
## $ social_support : int 0 1 0 0 0 1 0 0 0 0 ...
## $ depression : Factor w/ 5 levels "1","2","3","4",...: 2 1 2 2 1 2 2 1 1 5 ...
```

The data set contains 8 independent random variables with one dependent variable (depression) measured at 5 ordered levels. There were a total of 700 observations from the study.

```
# Histogram of Age_Participant
hist(dataset$age_participant, main = "Histogram of Age of the Caregiver", xlab = "Age")
```

A plot of the histogram shows that the age of the caregivers was normally distributed with an average age of 40 years.

Histogram of Age of the Caregiver

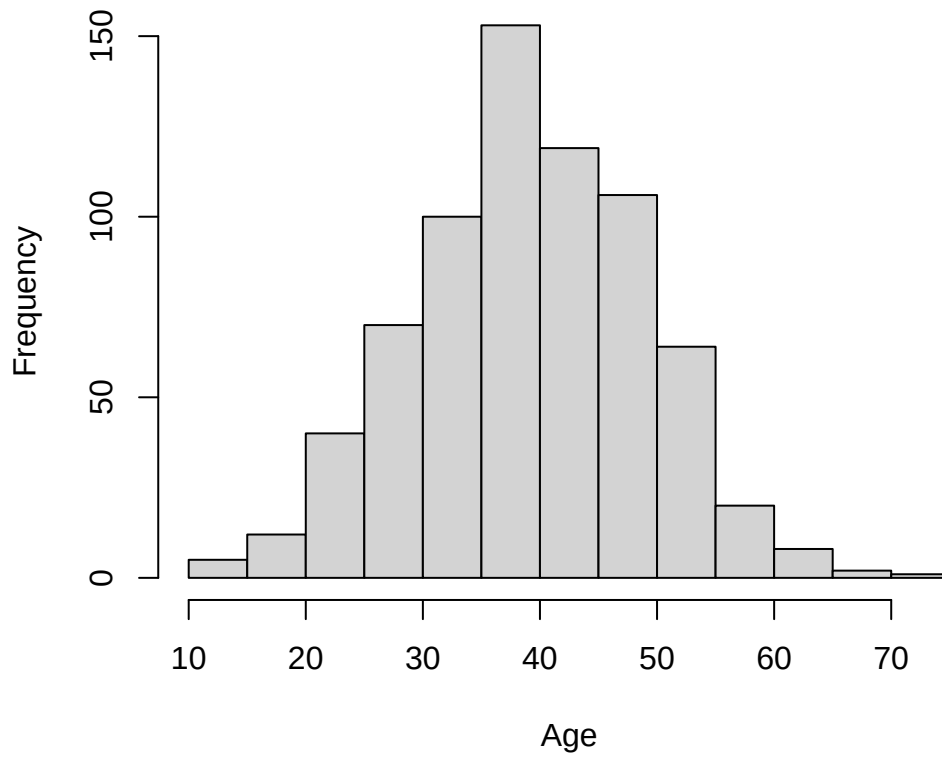
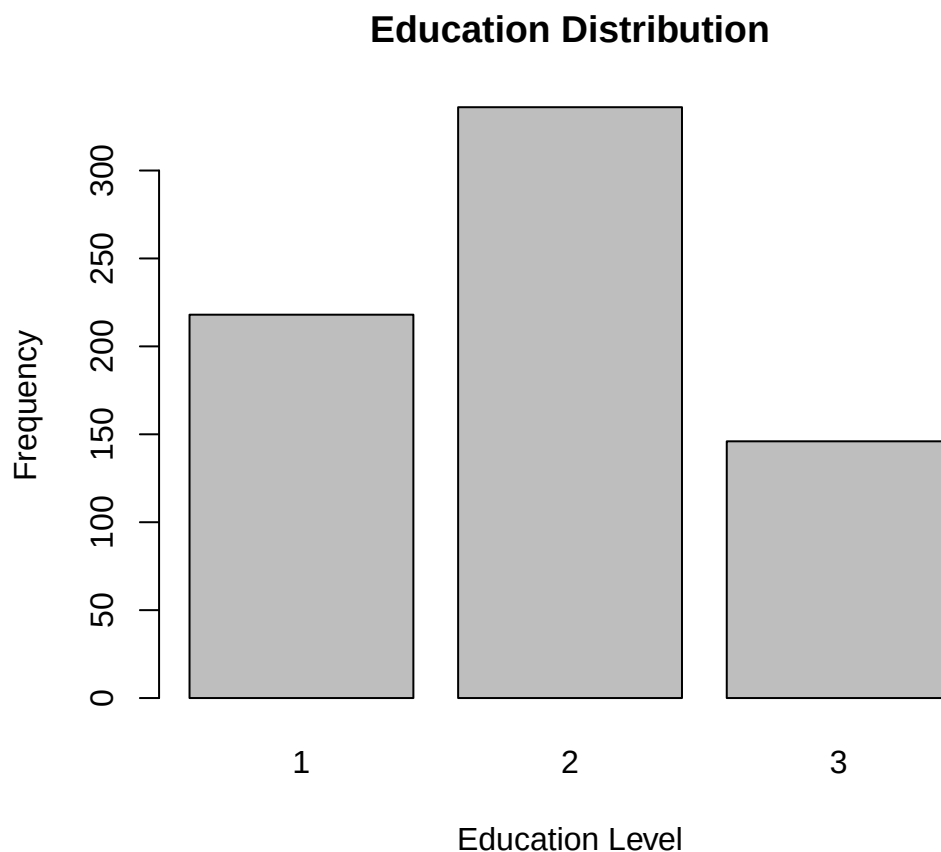


Figure 4.1: Histogram of age of caregivers

```
# Bar plot of Education  
barplot(table(dataset$education), main = "Education Distribution", xlab = "Education")
```



A plot of the histogram shows that the most of the caregivers on average had attained Secondary school education.

```
# Box plot of Income_Month by Depression Level
boxplot(dataset$income_ksh ~ dataset$depression, main = "Income by Depression Level",
```

```
# Correlation matrix
names(dataset)

## [1] "age_participant" "age_patient"      "sex_participant" "sex_patient"
## [5] "marital_status"  "education"         "income_ksh"      "social_support"
## [9] "depression"
```

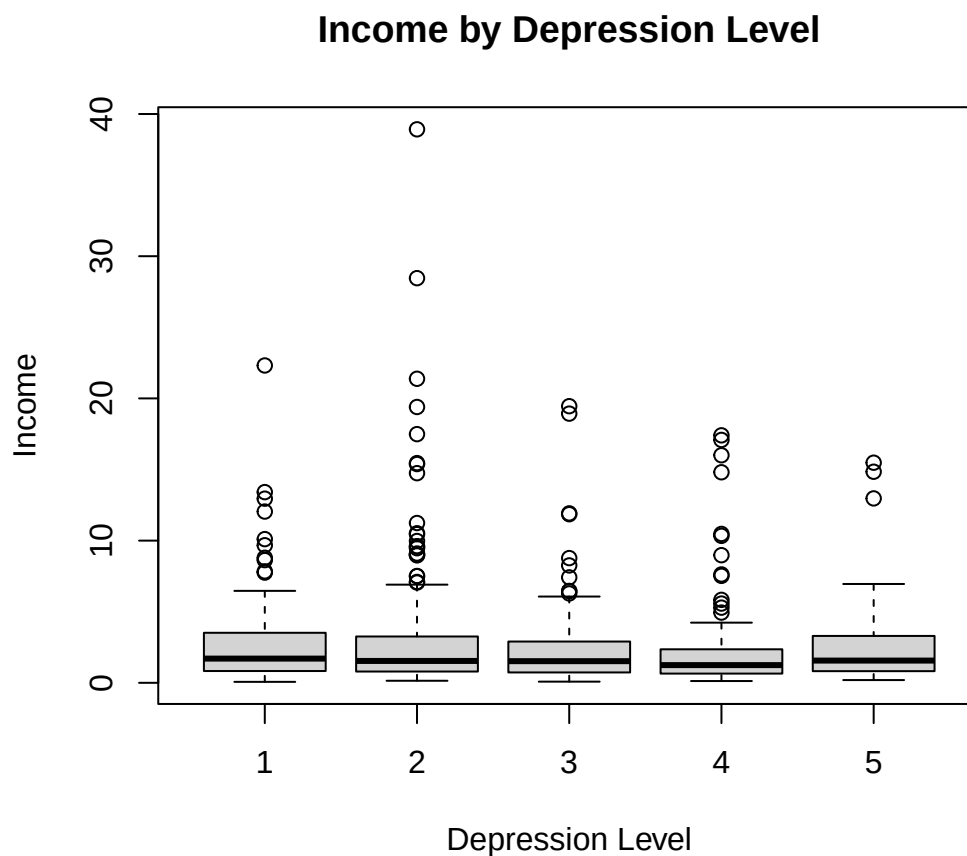


Figure 4.2: Box plot of income by depression level

```
cor_matrix <- cor(dataset[, c("age_participant", "age_patient", "income_ksh")])
print(cor_matrix)

##           age_participant age_patient  income_ksh
## age_participant      1.00000000  0.07172348 -0.04748347
## age_patient          0.07172348  1.00000000 -0.01814071
## income_ksh           -0.04748347 -0.01814071  1.00000000
```

The correlation matrix indicated a positive correlation between age of the care giver and the income the caregiver earns. It was noted that there is a negative correlation

between age of the patient and income earned by the caregiver.

```
# Pairwise scatter plot  
pairs(dataset[, c("age_participant", "age_patient", "income_ksh")])
```

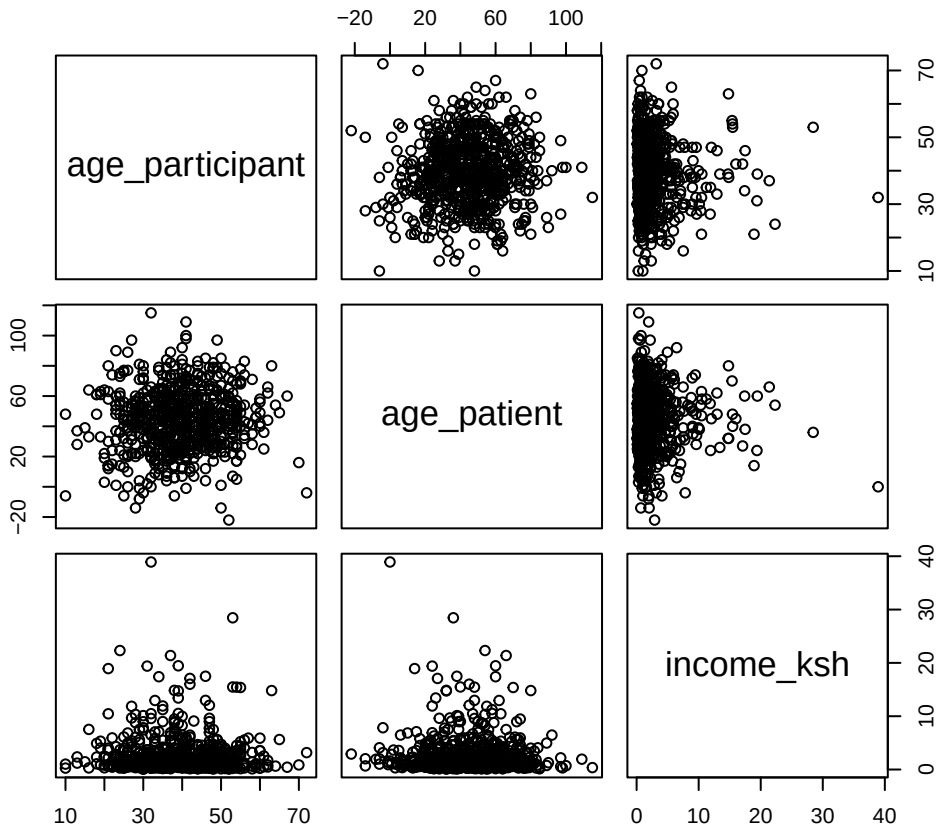


Figure 4.3: Pairwise scatter plot

The data forms ellipsoid clusters indicating that there is a relationship in the ages of the caregiver, age of patient and income of the caregiver.

The data met all the assumptions of ordinal logistic regression that is Proportional Odds Assumption, independence of Observations, Ordinality of the Dependent Variable and adequate Sample Size. Therefore the data was suitable to fit an ordinal

logistic regression to assess the effects of the age of patient, sex, marital status, education, income and age of care giver on the depression levels of the caregiver.

4.5 Ordinal Logistic Regression Model

```
# Fit an ordinal logistic regression model
model <- polr(depression ~ age_participant + age_patient +
              marital_status + education + income_ksh + social_support,
              data = df, Hess=TRUE)
```

4.5.1 Prediction of New Observations

Since the essence of a regression model is to make predictions of future outcomes, we build the prediction model as follows;

```
# Collect predictor variables for new observations (replace with actual values)
new_obs <- data.frame( age_participant=c(30,45,52),
                       age_patient =c(12,6,40),
                       marital_status =c(1,3,2),
                       education=c(1,3,2),
                       income_ksh =c(10,56,16),
                       social_support=c(0,1,0))

# Calculate the linear predictors for each threshold
linear_predictor <- model$zeta[1] * new_obs$age_participant +
  model$zeta[2] * new_obs$age_patient+
  model$zeta[3] * new_obs$marital_status+
  model$zeta[4] * new_obs$education+
  model$zeta[5] * new_obs$income_ksh+
  model$zeta[6] * new_obs$social_support

# Use the logistic CDF to obtain probabilities for each threshold
probabilities <- plogis(linear_predictor)

# Assign the predicted level based on the highest probability
```

```

predictions <- predict(model, newdata = new_obs, type = "probs")
# Print the predicted level
predictions

##           1           2           3           4           5
## 1 0.2157717 0.3511761 0.2232167 0.13248514 0.07735036
## 2 0.3646954 0.3673149 0.1550796 0.07428051 0.03862958
## 3 0.2273448 0.3560009 0.2177301 0.12622980 0.07269440

```

Proportional Odds Assumption

One of the assumptions underlying ordinal logistic (and ordinal probit) regression is that the relationship between each pair of outcome groups is the same. Ordinal logistic regression assumes that the coefficients that describe the relationship between, say, the lowest versus all higher categories of the response variable are the same as those that describe the relationship between the next lowest category and all higher categories, etc.

Thus, in order to assess the appropriateness of our model, we need to evaluate whether the proportional odds assumption is tenable.

```

sf <- function(y) {
  c('Y>=1' = qlogis(mean(y >= 1)),
    'Y>=2' = qlogis(mean(y >= 2)),
    'Y>=3' = qlogis(mean(y >= 3)),
    'Y>=4' = qlogis(mean(y >= 4)),
    'Y>=5' = qlogis(mean(y >= 5)))
}

(s <- with(df, summary(as.numeric(depression) ~ age_participant +
  age_patient + sex_participant +
  marital_status + education + income_ksh + social_support, fu

```

```

## as.numeric(depression)      N= 700
##
## +-----+-----+-----+-----+-----+-----+
## |           |           | N|Y>=1|   Y>=2|   Y>=3|   Y>=4|
## +-----+-----+-----+-----+-----+-----+
## |age_participant|           [10,34]|179|  Inf|1.182311|-0.21309322|-1.091177|
## |           |           [34,41]|201|  Inf|1.243194|-0.27029033|-1.392525|

```


##				[41,47)	146	Inf 1.783791 -0.13720112 -1.270463
##				[47,72]	174	Inf 1.452252 -0.06899287 -1.176777
##	+-----+-----+-----+-----+-----+-----+					
##		age_patient		[-22, 32)	177	Inf 1.436484 -0.12445417 -1.231101
##				[32, 46)	182	Inf 1.400088 -0.13205972 -1.299283
##				[46, 59)	174	Inf 1.241713 -0.27763174 -1.176777
##				[59,115]	167	Inf 1.439539 -0.18012617 -1.222226
##	+-----+-----+-----+-----+-----+-----+					
##		sex_participant		No	346	Inf 1.408113 -0.19716806 -1.354091
##				Yes	354	Inf 1.347895 -0.15852323 -1.121341
##	+-----+-----+-----+-----+-----+-----+					
##		marital_status		1	303	Inf 1.484037 -0.08586129 -1.221215
##				2	259	Inf 1.310787 -0.25621880 -1.220780
##				3	138	Inf 1.280934 -0.23293156 -1.280934
##	+-----+-----+-----+-----+-----+-----+					
##		education		1	218	Inf 1.523824 -0.14705342 -1.186168
##				2	336	Inf 1.317050 -0.21511138 -1.299283
##				3	146	Inf 1.310945 -0.13720112 -1.154182
##	+-----+-----+-----+-----+-----+-----+					
##		income_ksh		[0.0676, 0.744)	175	Inf 1.422399 -0.05715841 -1.121602
##				[0.7435, 1.536)	175	Inf 1.459319 -0.21800215 -1.249093
##				[1.5356, 3.117)	175	Inf 1.459319 -0.03428907 -1.152680
##				[3.1173,38.923]	175	Inf 1.184268 -0.40546511 -1.422399
##	+-----+-----+-----+-----+-----+-----+					
##		social_support		No	504	Inf 1.252763 -0.15906469 -1.196251
##				Yes	196	Inf 1.750698 -0.22543976 -1.329853
##	+-----+-----+-----+-----+-----+-----+					
##		Overall			700	Inf 1.377390 -0.17760827 -1.232671
##	+-----+-----+-----+-----+-----+-----+					

The table above displays the (linear) predicted values we got if we regressed our dependent variable on our predictor variables one at a time, without the parallel slopes assumption. To evaluate the parallel slopes assumption we ran a series of binary logistic regressions with varying cutpoints on the dependent variable and checking the equality of coefficients across cutpoints.

We thus relaxed the parallel slopes assumption to check its tenability. To accomplish this, we transformed the original, ordinal, dependent variable into a new, binary,

dependent variable which is equal to zero if the original, ordinal dependent variable (depression) is less than some value a , and 1 if the ordinal variable is greater than or equal to a for all the 4 levels of depression. From the results we obtained, we found that relationship between each pair of outcome groups is the same suggesting that the proportional odds assumption was met.

4.6 Data Interpretation and Discussions

```
# Print the model summary
#model
summary(model)

## Call:
## polr(formula = depression ~ age_participant + age_patient + marital_status +
##       education + income_ksh + social_support, data = df, Hess = TRUE)
##
## Coefficients:
##               Value Std. Error t value
## age_participant -0.0052639   0.006954  0.7569
## age_patient      -0.0008122   0.003406  0.2385
## marital_status    0.0758508   0.089496 -0.8475
## education         0.0496053   0.095367 -0.5201
## income_ksh        0.0133679   0.018344 -0.7287
## social_support    -0.0563137   0.149247  0.3773
##
## Intercepts:
##      Value   Std. Error t value
## 1|2 -1.3819   0.3918    -3.5275
## 2|3  0.1779   0.3877     0.4589
## 3|4  1.2344   0.3906     3.1606
## 4|5  2.3874   0.4035     5.9168
##
## Residual Deviance: 215.684
## AIC: 235.684
```

Let $\mathbf{Y}$ be an ordinal outcome with j categories.

Then $P(Y \leq j)$ is the cumulative probability of $\mathbf{Y}$ less than or equal to a specific category $j = 1, \dots, j - 1$.

The odds of being less than or equal a particular category can be defined as $\frac{P(Y \leq j)}{P(Y > j)}$

For $j = 1, \dots, j - 1$ since $P(Y > J) = 0$ and dividing by zero is undefined.

The log odd also known as the *logit*, so that

$\log \frac{P(Y \leq j)}{P(Y > j)} = \text{logit}(P(Y \leq j))$ The estimated logit model is;

$\text{logit}(P(Y \leq 2)) = -1.38 + 0.005 * \text{age_participant} + 0.0008 * \text{age_patient} - 0.076 * \text{marital_status} - 0.05 * \text{education} - 0.013 * \text{income} + 0.056 * \text{social_support}$

$\text{logit}(P(Y \leq 2)) = 0.18 + 0.005 * \text{age_participant} + 0.0008 * \text{age_patient} - 0.076 * \text{marital_status} - 0.05 * \text{education} - 0.013 * \text{income} + 0.056 * \text{social_support}$

$\text{logit}(P(Y \leq 3)) = 1.23 + 0.005 * \text{age_participant} + 0.0008 * \text{age_patient} - 0.076 * \text{marital_status} - 0.05 * \text{education} - 0.013 * \text{income} + 0.056 * \text{social_support}$

$\text{logit}(P(Y \leq 4)) = 2.39 + 0.005 * \text{age_participant} + 0.0008 * \text{age_patient} - 0.076 * \text{marital_status} - 0.05 * \text{education} - 0.013 * \text{income} + 0.056 * \text{social_support}$

4.6.1 Model Statistics

```
#model statistics
model$coefficients

## age_participant      age_patient marital_status      education      income_ksh
##      0.0052639339      0.0008122355      -0.0758508009      -0.0496053286      -0.0133679199
## social_support
##      0.0563137072

model$zeta

##          1|2          2|3          3|4          4|5
## -1.3819499  0.1779383  1.2344467  2.3874338

model$deviance

## [1] 215.684

model$df.residual

## [1] 690
```

```

model$edf

## [1] 10

model$call

## polr(formula = depression ~ age_participant + age_patient + marital_status +
##       education + income_ksh + social_support, data = df, Hess = TRUE)

model$convergence

## [1] 0

```

The thresholds are denoted as 1|2, 2|3, 3|4, and 4|5. The zeta values indicate the direction and strength of the relationship between the predictor variables and the odds of transitioning from one level to the next. Example, a zeta value of 2.39 for the 4|5 threshold suggests that as the predictor variables increase, the odds of transitioning from level 1 to level 2 increases.

The deviance value of 215.684 suggests that the model fit the data very well, as it is relatively low.

The df.residual value of 690 indicates the degrees of freedom for the residuals.

The edf value of 10 represents the effective degrees of freedom, which takes into account the ordinal nature of the response variable.

The convergence value of 0 suggests that the model converged successfully during the estimation process

.
Call, recalls the type of model we ran, and options we specified.

Also, we recall output coefficient table which shows value of each coefficient, standard errors, and t value. This represents the ratio of the coefficient to its standard error.

```

## store table
(ctable <- coef(summary(model)))

```

##		Value	Std. Error	t value
##	age_participant	0.0052639339	0.006954489	0.7569117
##	age_patient	0.0008122355	0.003405856	0.2384821
##	marital_status	-0.0758508009	0.089495806	-0.8475347
##	education	-0.0496053286	0.095367414	-0.5201497
##	income_ksh	-0.0133679199	0.018344140	-0.7287297
##	social_support	0.0563137072	0.149247258	0.3773182
##	1 2	-1.3819499210	0.391768796	-3.5274630
##	2 3	0.1779383054	0.387722932	0.4589316
##	3 4	1.2344467404	0.390578115	3.1605630
##	4 5	2.3874338238	0.403501520	5.9167902

Next we see the estimates for the four intercepts. The intercepts indicate where the latent variable is cut to make the three groups that we observe in our data.

```
## calculate and store p values
p <- pnorm(abs(ctable[, "t value"]), lower.tail = FALSE) * 2
```

Finally, we see the residual deviance, $-2 * \text{Log Likelihood}$ of the model as well as the AIC. Both the deviance and AIC are useful in model comparison.

```
## combined table
(ctable <- cbind(ctable, "p value" = p))
```

##		Value	Std. Error	t value	p value
##	age_participant	0.0052639339	0.006954489	0.7569117	4.491028e-03
##	age_patient	0.0008122355	0.003405856	0.2384821	8.115072e-01
##	marital_status	-0.0758508009	0.089495806	-0.8475347	3.966972e-03
##	education	-0.0496053286	0.095367414	-0.5201497	6.029593e-03
##	income_ksh	-0.0133679199	0.018344140	-0.7287297	4.661670e-03
##	social_support	0.0563137072	0.149247258	0.3773182	7.059371e-03
##	1 2	-1.3819499210	0.391768796	-3.5274630	4.195624e-04
##	2 3	0.1779383054	0.387722932	0.4589316	6.462833e-03
##	3 4	1.2344467404	0.390578115	3.1605630	1.574645e-03
##	4 5	2.3874338238	0.403501520	5.9167902	3.282851e-09

```

#confidence intervals
(ci <- confint(model))

## Waiting for profiling to be done...

##              2.5 %      97.5 %
## age_participant -0.008351804 0.018923591
## age_patient     -0.005870204 0.007487586
## marital_status  -0.251513448 0.099457036
## education       -0.236714705 0.137291088
## income_ksh      -0.049542705 0.022781242
## social_support  -0.236402674 0.348893313

```

We also calculated confidence intervals for the parameter estimates. We obtained the confidence intervals by profiling the likelihood function. It was noted that profiled Confidence Intervals are not symmetric. Since our 95% CI did not cross 0, then we conclude the parameter estimates are statistically significant.

For a unit increase in the age of a caregiver, the expected change of the value of depression in log odds scale is 0.005 given all the other variables are held constant.

For a unit increase in the age of the patient, the expected change of the value of depression in the log odds scale is 0.0008 given all the other variables are held constant.

For income, a one unit increase in income, we would expect a 0.013367 decrease in the expected value of depression in the log odds scale, given that all of the other variables in the model are held constant.

A unit increase in education(i.e from primary to secondary), we would expect a 0.05 decrease in the value of depression in the log odds scale, given that all of the other variables in the model are held constant.

A one unit increase in marital status, we would expect a 0.076 decrease in the value of depression in the log odds scale, given that all of the other variables in the model are held constant.

A one unit increase in social support, the expected change in the value of depression in log odds scale is 0.06, given that all of the other variables in the model are held

constant..

4.7 Odds Ratio

```
## odds ratios
exp(coef(model))

## age_participant      age_patient  marital_status      education      income_ksh
##      1.0052778      1.0008126      0.9269545      0.9516049      0.9867210
## social_support
##      1.0579295

## OR and CI
exp(cbind(OR = coef(model), ci))

##              OR      2.5 %   97.5 %
## age_participant 1.0052778 0.9916830 1.019104
## age_patient     1.0008126 0.9941470 1.007516
## marital_status  0.9269545 0.7776230 1.104571
## education       0.9516049 0.7892164 1.147162
## income_ksh      0.9867210 0.9516645 1.023043
## social_support  1.0579295 0.7894627 1.417498
```

- **Age Caregiver:**
One unit increase in age of caregiver increases the odds of being depressed by 0.5% assuming all other variables are held constant.
- **Age Patient:**
One unit increase in age of the patient increases the odds of the caregiver being depressed by 0.08% assuming all other variables are held constant.
- **Marital Status:** Caregivers who are married (compared to those who are not) have lower odds of being in a higher depression level by approximately 8.4%, assuming all other variables are held constant.
- **Education:**
One-unit increase in Education(say from primary to secondary) reduces the

odds of the caregiver being depressed by 5.9%, assuming all other variables are held constant.

- **Household Income:**

One-unit increase in Household income reduces the odds of being in higher depression level by 1.4% assuming all other variables are held constant.

- **Social Support:**

Caregivers involved in social support groups (compared to those not involved) have higher odds of being in a higher depression level by approximately 5.79%, assuming all other variables are held constant.

4.7.1 Hypothesis Testing

Likelihood Ratio Test:

```
# 1. Likelihood Ratio Test
null_model <- polr(depression ~ 1, data = dataset)
# Null model with only intercept
likelihood_ratio <- anova(null_model, model)
# Likelihood ratio test
likelihood_ratio

## Likelihood ratio tests of ordinal regression models
##
## Response: depression
##
## 1
## 2 age_participant + age_patient + marital_status + education + income_ksh + social
##   Resid. df Resid. Dev   Test    Df LR stat.   Pr(Chi)
## 1      696    218.060
## 2      690    215.684 1 vs 2      6 2.375745 0.008821064

p_value_lr <- likelihood_ratio$Pr[2]
# p-value for the likelihood ratio test
p_value_lr

## [1] 0.008821064
```


The likelihood ratio test compares the fit of the fitted model with a null model that includes only the intercept. This helps determine whether the independent variables significantly improve the model fit. The likelihood ratio test is significant suggesting that the independent variables collectively have a significant effect on the odds of being in a higher depression level.

CHAPTER FIVE

CONCLUSIONS

5.1 Introduction

This chapter presents the summary and the conclusions of the study in the subsections that follow.

5.2 Summary

In summary, Exploratory data analysis on the dataset was performed and found that the data met all the assumptions of ordinal logistic regression that is Proportional Odds Assumption, independence of Observations, Ordinality of the Dependent Variable and adequate Sample Size. Therefore, the data was suitable to fit an ordinal logistic regression to assess the effects of the age of patient, marital status, education, income and age of care giver on the depression levels of the cancer caregiver. Lastly, significant depression predictors were identified, estimation and interpretation of coefficients of factors of depression were done and determination and interpretation of the odds ratios were done.

5.3 Conclusions

In conclusion we found that the age of caregiver, Household income, level of education, Marital status of the caregiver and social support are significant in determining the depression severity of cancer caregivers while the age of the patient was found to be insignificant.

The ordinal logistic regression model fits our dataset since the likelihood ratio test is significant suggesting that the independent variables collectively have a significant effect on the odds of being in a higher depression level.

5.4 Recommendation

Based on the findings of this research project, several evidence-based recommendations can be made to inform the development of targeted interventions for depression among cancer caregivers in rural settings:

1. For easier availability of cancer caregivers data in Kenya, Kenya National Bureau of Statistics to conduct research and store data on Depression for future research.
2. Given the role being performed by cancer caregivers, the government through state departments should develop interventions to provide tailored support to cancer caregivers for better quality life of the patients.
3. The level of income emerged as an important factor in the ordinal logistic regression analysis. Interventions should address the financial challenges faced by caregivers in rural areas, such as limited job opportunities or transportation costs. Offering financial assistance programs or connecting caregivers with available resources can help alleviate financial stress and improve their mental well-being.

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